King Fahd University of Petroleum and Minerals

Department of Mathematics

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Math 101 Final Exam Term 221

Original exam date: not printed. Practice version: questions and choices shuffled.

Net time allowed: 120 minutes

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1. If $f(x) = \dfrac{2x^3 - 3x + 5}{x^2 + x}$, then an equation of the slant asymptote for the graph of f is

(a) $y = 3x - 2$

(b) $y = 2x - 2$

(c) $y = 2x - 3$

(d) $y = x + 3$

(e) $y = 3x + 1$

2. The function

$$f(x) = \frac{2}{x \ln x}$$

has

(a) no vertical and one horizontal asymptote

(b) two vertical and no horizontal asymptotes

(c) one vertical and no horizontal asymptote

(d) one vertical and one horizontal asymptotes

(e) two vertical and one horizontal asymptotes

3. $\lim_{x \to -\infty} \left(2x + \sqrt{4x^2 - x}\right) =$

(a) $\dfrac{1}{4}$

(b) 0

(c) $\dfrac{1}{2}$

(d) 2

(e) does not exist

4. If $x^2y - 4x = 5$, then $\dfrac{d^2y}{dx^2}$ at the point $(-1, 1)$ is equal to

(a) 38

(b) 22

(c) 14

(d) 6

(e) 24

5. If $f''(x) = \dfrac{2}{x^2}$, $f'(1) = 4$ and $f(1) = 3$, then $2f(2) - f(4) =$

(a) -3

(b) $8\ln 2 + 3$

(c) $-4\ln 2 + 21$

(d) $4\ln 2 + 18$

(e) 15

6. The area of the largest rectangle that can be inscribed in the region bounded by the x-axis, the y-axis and the line $y = 3 - \dfrac{1}{2}x$ is equal to

(a) $\dfrac{9}{2}$

(b) 9

(c) 6

(d) 3

(e) $\dfrac{7}{2}$

7. $\lim_{x\to 3^+} [4(x-3)]^{(x-3)} =$

(a) 4

(b) 2

(c) 3

(d) 5

(e) 1

8. The sum of all values of x at which the function $f(x) = \dfrac{-4x}{\sqrt{2x-1}}$ has a horizontal tangent line is

(a) $\dfrac{3}{2}$

(b) $-\dfrac{3}{2}$

(c) 0

(d) -1

(e) 1

9. Using the tangent line approximation of $f(x) = \sqrt{x+9}$ at the point $(0, 3)$, $\sqrt{9.03}$ can be approximated as

(a) 3.005

(b) 3.003

(c) 3.004

(d) 3.001

(e) 3.002

10. The sum of all values of x at which the tangent lines to the graph of $y = \dfrac{2x-1}{3x+1}$ are parallel to the line $4x - 5y + 1 = 0$ is

(a) $-\dfrac{4}{5}$

(b) $-\dfrac{1}{4}$

(c) $-\dfrac{2}{3}$

(d) $-\dfrac{3}{5}$

(e) -1

11. Let

$$f(x) = \begin{cases} x + b & \text{if } 0 < x < 2 \\ ax^2 + 1 & \text{if } 2 \leq x \leq 4 \end{cases}$$

If f is continuous at $x = 2$, then $4a - b =$

(a) 2

(b) -1

(c) 4

(d) 1

(e) 3

12. If $f''(x)$ is a continuous function, then

$$\lim_{h\to 0} \frac{3f(x+4h) - 7f(x) + 4f(x-3h)}{6h^2} =$$

(a) $f''(x)$

(b) $3f'(x)$

(c) $7f''(x)$

(d) $3f''(x) + 4f'(x)$

(e) $4f''(x)$

13. The absolute maximum of the function $f(x) = \dfrac{x^2}{x-2}$ on $[-2, 1]$ is

(a) 2

(b) 8

(c) -1

(d) 0

(e) 1

14. If $\lim_{x\to-\infty} \left(\dfrac{3x-2}{\sqrt{4x^2+5}}\right) = L_1$ and $\lim_{x\to\infty} \left(\dfrac{3x-2}{\sqrt{4x^2+5}}\right) = L_2$, then $L_1 - L_2 =$

(a) ∞

(b) $-\dfrac{3}{2}$

(c) $-\infty$

(d) 0

(e) -3

15. $\lim_{x\to 0} \frac{1-\cos x}{x^2} =$

(a) 1

(b) 0

(c) $\dfrac{1}{2}$

(d) ∞

(e) $-\infty$

16. If $h(x) = xf(x)g(x)$, with $f(1) = 2$, $g(1) = 2$, $f'(1) = 1$ and $g'(1) = 1$, then $h'(1) =$

(a) 8

(b) 6

(c) 4

(d) 1

(e) 5

17. If $f(x) = \frac{1}{2}x + 2$, then the largest number $\delta > 0$ such that $|f(x) - 3| < 0.45$ whenever $0 < |x - 2| < \delta$ is equal to

(a) $\frac{12}{10}$

(b) $\frac{11}{10}$

(c) 1

(d) $\frac{9}{10}$

(e) $\frac{8}{10}$

18. If $f(x) = \cosh(\ln x)$, then $f'(1/5) =$

(a) -26

(b) -1

(c) 24

(d) -12

(e) -10

19. If $y = 0$ and $y = 5$ are two horizontal asymptotes of the function

$$f(x) = \frac{ae^x + be^{-x}}{2e^x + 3e^{-x}}, \qquad b > 0,$$

then $5a + b =$

(a) 5

(b) 15

(c) 10

(d) 30

(e) 6

20. If $\lim_{x\to 0} \dfrac{\sqrt{b-ax}-2}{x} = 3$, then $a+b=$

(a) 8

(b) 0

(c) -8

(d) 6

(e) -6

21. If M and N are the numbers of relative maximum points and relative minimum points of the function $f(x) = 5x^{4/5}(x-1)^2$ respectively, then $2M+N=$

(a) 4

(b) 3

(c) 5

(d) 2

(e) 6

22. The function $f(x) = x\sinh(x-1) - \cosh(x-1)$ has

(a) one relative minimum

(b) one relative maximum

(c) no relative minimum

(d) two critical points

(e) two relative extrema

23. The sum of all x-coordinates of the points at which the curve

$$x^2 - 5x + 3y^3 + 6 = 0$$

has vertical tangent line is

(a) 3

(b) 2

(c) 5

(d) 6

(e) 4

24. The sum of all values of x at which the graph of $f(x) = \dfrac{x}{e \ln x}$ has a point of inflection is

(a) e^2

(b) e^{-1}

(c) e^{-2}

(d) e

(e) $\ln 2$

25. If $(-12, 0)$ is the largest open interval over which the graph of

$$f(x) = x^4 + ax^3$$

is concave downward, then $a =$

(a) 27

(b) 12

(c) 24

(d) 15

(e) 19

26. If $f(x) + f''(x) = 1$ for all x, and $F(x)$ is an anti-derivative of $f(x)$ such that $F(0) = -1$, $F(1) = 2$ and $f'(0) = 2$, then $f'(1) =$

(a) 0

(b) 1

(c) -1

(d) -2

(e) 2

27. If an open box of maximum volume is to be made from a square piece of material, 30 cm on a side, by cutting equal squares from the corners and turning up the sides, then the sum of the dimensions of the box is

(a) 42 cm

(b) 45 cm

(c) 40 cm

(d) 30 cm

(e) 48 cm

28. The radius of a spherical balloon is measured as 8 cm, with a possible error of 0.02 cm. Using differentials, the percent error in computing the volume of the sphere is (Hint: $V = \frac{4}{3}\pi r^3$)

(a) 0.50 %

(b) 0.05 %

(c) 0.45 %

(d) 0.75 %

(e) 0.25 %

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	Long division: $f = 2x - 2 + \frac{5-x}{x^2+x}$, so $y = 2x - 2$.
2	**(e)**	Vertical at $x = 0$ and $x = 1$; horizontal $y = 0$.
3	**(a)**	Conjugate: $\frac{x}{2x-\sqrt{4x^2-x}} = \frac{1}{2+\sqrt{4-1/x}} \to \frac{1}{4}$.
4	**(b)**	$y = 5x^{-2} + 4x^{-1}$, $y'' = 30x^{-4} + 8x^{-3}$, at $x = -1$: 22.
5	**(a)**	$f' = 6 - \frac{2}{x}$, $f = 6x - 2\ln x - 3$; $2f(2) - f(4) = -3$.
6	**(a)**	$A = 3x - \frac{1}{2}x^2$, max at $x = 3$: $A = 3 \cdot \frac{3}{2} = \frac{9}{2}$.
7	**(e)**	$\ln y = \frac{\ln 4t}{1/t} \to 0$ by L'Hopital, so the limit is $e^0 = 1$.
8	**(e)**	$f' = 0 \iff 4 - 4x = 0$, so $x = 1$ (only value in the domain).
9	**(a)**	$L(x) = 3 + \frac{x}{6}$, $x = 0.03$: $\sqrt{9.03} \approx 3.005$.
10	**(c)**	$y' = \frac{5}{(3x+1)^2} = \frac{4}{5}$ gives $x = \frac{1}{2}, -\frac{7}{6}$; sum $-\frac{2}{3}$.
11	**(d)**	$2 + b = 4a + 1$, so $4a - b = 1$.
12	**(c)**	L'Hopital twice: $\frac{48f''+36f''}{12} = 7f''(x)$.
13	**(d)**	$f' = \frac{x(x-4)}{(x-2)^2}$; compare $f(-2) = -1$, $f(0) = 0$, $f(1) = -1$: max 0.
14	**(e)**	$L_1 = -\frac{3}{2}$, $L_2 = \frac{3}{2}$, so $L_1 - L_2 = -3$.

Q	Ans	Brief explanation
15	**(c)**	L'Hopital: $\frac{\sin x}{2x} \to \frac{1}{2}$.
16	**(a)**	$h' = fg + xf'g + xfg'$, so $h'(1) = 4 + 2 + 2 = 8$.
17	**(d)**	$\lvert f(x) - 3\rvert = \frac{1}{2}\lvert x - 2\rvert < 0.45$, so $\delta = 0.9 = \frac{9}{10}$.
18	**(d)**	$f = \frac{1}{2}(x + \frac{1}{x})$, $f' = \frac{1}{2}(1 - \frac{1}{x^2})$, so $f'(\frac{1}{5}) = -12$.
19	**(b)**	Limits $\frac{a}{2}$ and $\frac{b}{3}$; $b > 0$ forces $a = 0$, $b = 15$.
20	**(c)**	$b = 4$; conjugate gives $-a/4 = 3$, $a = -12$; $a + b = -8$.
21	**(a)**	$f' = x^{-1/5}(x - 1)(14x - 4)$: one maximum, two minima, so $2M + N = 4$.
22	**(a)**	$f' = x\cosh(x - 1)$: sign change $- \to +$ at $x = 0$ only, one relative minimum.
23	**(c)**	$y' = \frac{5-2x}{9y^2}$; $y = 0$ gives $x^2 - 5x + 6 = 0$, $x = 2, 3$, sum 5.
24	**(a)**	$f'' = \frac{1}{e}\frac{2-\ln x}{x\ln^3 x}$ changes sign only at $x = e^2$.
25	**(c)**	$f'' = 6x(2x + a) < 0$ on $(-\frac{a}{2}, 0)$; $-\frac{a}{2} = -12$ gives $a = 24$.
26	**(a)**	$F + f' = x + C$; $x = 0$ gives $C = 1$; $x = 1$: $2 + f'(1) = 2$, so $f'(1) = 0$.
27	**(b)**	$V = x(30 - 2x)^2$, $V' = 0$ at $x = 5$; dimensions $20, 20, 5$, sum 45.
28	**(d)**	$\frac{dV}{V} = \frac{3\,dr}{r} = \frac{0.06}{8} = 0.0075$, i.e. 0.75%.